Solstice® LBA

000000012786

Version 1.12 Issuing date 11/02/2011 Revision Date 08/24/2021 Print Date 01/19/2022

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product information

Trade name	:	Solstice® LBA
Number	:	000000012786
Recommended use of the chemical and restrictions on use	:	Foam blowing agent
Manufacturer or supplier's details	:	Honeywell International Inc. 115 Tabor Road Morris Plains, NJ 07950-2546 USA
For further information, please contact:	:	800-522-8001 In China: 021-28942149 (Monday-Friday, 9:00am-5:00pm)
In case of emergency call	:	**Medical: 1-800-498-5701 or +1-303-389-1414**
	:	**Transportation (CHEMTREC): 1-800-424-9300 or +1-703-527-3887** **In China: 4001-204937 or +86 532-83889090**

2. HAZARDS IDENTIFICATION

Classification of the substance or mixture

Classification of the substance or mixture	:	Gases under pressure, Liquefied gas Short-term (acute) aquatic hazard, Category 3

GHS Label elements, including precautionary statements

Symbol(s) :



Signal word	:	Warning
Hazard statements	:	Contains gas under pressure; may explode if heated. Harmful to aquatic life.
Precautionary statements	:	**Prevention:** Avoid release to the environment. **Storage:** Protect from sunlight. Store in a well-ventilated place.

Disposal:
Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification : Causes asphyxiation in high concentrations. The victim will not realize that he/she is suffocating.
May cause cardiac arrhythmia.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical nature : Substance

Chemical name	CAS-No.	Concentration
trans-1-Chloro-3,3,3-trifluoropropene trans-1-Chloro-3,3,3-trifluoropropene	102687-65-0	>99.00 %

4. FIRST AID MEASURES

Inhalation : Remove to fresh air.
If not breathing, give artificial respiration.
If breathing is difficult, give oxygen.
Use oxygen as required, provided a qualified operator is present.
Call a physician.

Skin contact : After contact with skin, wash immediately with plenty of water.
If symptoms persist, call a physician.
Take off all contaminated clothing immediately.
Wash contaminated clothing before re-use.

Eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.
Call a physician if irritation develops or persists.

Ingestion : If victim is fully conscious, give a cupful of water.
Do not induce vomiting without medical advice.
Never give anything by mouth to an unconscious person.
Call a physician immediately.

Notes to physician : Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	The product is not flammable. Water spray Carbon dioxide (CO2) Dry chemical Foam
Specific hazards during firefighting	:	This product is not flammable at ambient temperatures and atmospheric pressure. However, this material can ignite when mixed with air under pressure and exposed to strong ignition sources. Container may rupture on heating. Cool closed containers exposed to fire with water spray. Do not allow run-off from fire fighting to enter drains or water courses. Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing. Exposure to decomposition products may be a hazard to health. Fire may cause evolution of: Hydrogen fluoride Hydrogen chloride gas Carbon oxides Halogenated compounds Carbonyl halides
Special protective equipment for firefighters	:	Wear full protective clothing and self-contained breathing apparatus.
Further information	:	Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Immediately evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Wear personal protective equipment. Unprotected persons must be kept away. Ventilate the area. Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing. Avoid accumulation of vapours in low areas. Unprotected personnel should not return until air has been tested and determined safe. Ensure that the oxygen content is >= 19.5%.
Environmental precautions	:	Do not flush into surface water or sanitary sewer system. Prevent further leakage or spillage if safe to do so. Prevent spreading over a wide area (e.g. by containment or oil

barriers).

Methods and materials for containment and cleaning up	:	Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

7. HANDLING AND STORAGE

Handling

Precautions for safe handling	:	Handle with care. Do not use in areas without adequate ventilation. Do not breathe vapours or spray mist. Avoid contact with skin, eyes and clothing. Follow all standard safety precautions for handling and use of compressed gas cylinders. Use authorized cylinders only. Protect cylinders from physical damage. Do not puncture or drop cylinders, expose them to open flame or excessive heat. Do not pierce or burn, even after use. Do not spray on a naked flame or any incandescent material. Do not remove screw cap until immediately ready for use. Always replace cap after use.
Advice on protection against fire and explosion	:	Can form a combustible mixture with air at pressures above atmospheric pressure. Keep product and empty container away from heat and sources of ignition.

Storage

Conditions for safe storage, including any incompatibilities	:	Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 55 °C. Keep containers tightly closed in a dry, cool and well-ventilated place. Storage rooms must be properly ventilated. Ensure adequate ventilation, especially in confined areas. Protect cylinders from physical damage. Store away from incompatible substances.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value	Control parameters	Update	Basis
trans-1-Chloro-3,3,3-trifluoropropene trans-1-Chloro-3,3,3-trifluoropropene	102687-65-0	TWA : Time weighted average	(800 ppm)	2014	WEEL:OARS - Workplace Environmental Exposure Level (WEEL) Guides

		TWA : Time weighted average	(800 ppm)	2013	Honeywell:Limit established by Honeywell International Inc.

Appropriate engineering controls

Use with local exhaust ventilation.
Perform filling operations only at stations with exhaust ventilation facilities.

Individual protection measures, such as personal protective equipment

Respiratory protection : In case of insufficient ventilation wear suitable respiratory equipment.
Wear a positive-pressure supplied-air respirator.
For rescue and maintenance work in storage tanks use self-contained breathing apparatus.
Use NIOSH approved respiratory protection.

Hand protection : Impervious gloves
Gloves must be inspected prior to use.
Replace when worn.

Eye protection : Wear as appropriate:
Safety glasses with side-shields
Safety goggles

Skin and body protection : Wear as appropriate:
Solvent-resistant gloves
Solvent-resistant apron and boots
If splashes are likely to occur, wear:
Protective suit

Hygiene measures : Handle in accordance with good industrial hygiene and safety

practice.
Avoid contact with skin, eyes and clothing.
Do not breathe vapours or spray mist.
Ensure adequate ventilation, especially in confined areas.
Remove and wash contaminated clothing before re-use.
Contaminated work clothing should not be allowed out of the workplace.
Keep working clothes separately.
Wash hands before breaks and immediately after handling the product.

Protective measures : Ensure that eyewash stations and safety showers are close to the workstation location.
Do not breathe vapours or spray mist.
Avoid contact with skin, eyes and clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	liquid, clear
Colour	:	colourless
Odour	:	slight
Melting point/range	:	< -90 °C Method: OECD Test Guideline 102
Boiling point/boiling range	:	19 °C Method: OECD Test Guideline 103
Flash point	:	Method: ISO 2719 Note: Not applicable
Flammability	:	The product is not flammable. Method: Flammability (gases)
Lower explosion limit	:	Note: None
Upper explosion limit	:	Note: None
Vapour pressure	:	1,516 hPa at 30 °C(86 °F)
Vapour density	:	Note: (Air = 1.0), not determined

Density	:	1.27 g/cm3
Water solubility	:	1.90 g/l at 20 °C Method: OECD Test Guideline 105
Partition coefficient: n-octanol/water	:	log Pow: 2.2 at 25 °C
Ignition temperature	:	380 °C at 986.8 - 1,035.9 hPa Method: DIN 51794
Decomposition temperature	:	> 250 °C
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.
Molecular weight	:	130.5 g/mol

10. STABILITY AND REACTIVITY

Chemical stability	:	Stable under recommended storage conditions.
Possibility of hazardous reactions	:	Polymerisation can occur.
Conditions to avoid	:	Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 55 °C. Can form a combustible mixture with air at pressures above atmospheric pressure. Do not mix with oxygen or air above atmospheric pressure.
Incompatible materials to avoid	:	Strong oxidizing agents Finely divided metal powders such as aluminum, magnesium, or zinc.
Hazardous decomposition products	:	Halogenated compounds Carbon oxides Hydrogen fluoride Carbonyl halides Hydrogen chloride gas

11. TOXICOLOGICAL INFORMATION

Acute inhalation toxicity	:	LC50: 120000 ppm

Exposure time: 4 h
Species: Rat

Skin corrosion/irritation : Species: Rabbit
Result: No skin irritation
Classification: Not classified as a skin irritant in animal testing.
Method: OECD Test Guideline 404
Exposure time: 4 h

Respiratory sensitisation : Note: No data available

Skin sensitisation : Result: Does not cause skin sensitisation.
Classification: Patch test on human volunteers did not demonstrate sensitisation properties.

Repeated dose toxicity : Species: Rat
Application Route: Inhalation
Exposure time: 4 Weeks
NOEL: 4500 ppm
Note: Subacute toxicity

Germ cell mutagenicity : Test Method: Mutagenicity (Salmonella typhimurium - reverse mutation assay)
Metabolic activation: with and without metabolic activation
Result: negative

: Test Method: Mutagenicity (Escherichia coli - reverse mutation assay)
Metabolic activation: with and without metabolic activation
Result: negative
Method: OECD Test Guideline 471

: Test Method: Chromosome aberration test in vitro
Cell type: Human lymphocytes
Result: negative
Method: OECD Test Guideline 473

: Species: Rat
Cell type: Bone marrow
Method: Mutagenicity (micronucleus test)
Result: negative

: Test Method: Unscheduled DNA synthesis
Species: Rat
Result: negative

: Species: Mouse
Cell type: Bone marrow
Method: Mutagenicity (micronucleus test)
Result: negative

Carcinogenicity : Note: No data available

Reproductive toxicity : Species: Rabbit
Note: No-observed-effect level - 15,000 ppm

: Species: Rat
Note: No-observed-effect level - 10,000 ppm

Teratogenicity : Species: Rabbit
Note: No-observed-effect level - 15,000 ppm

: Species: Rat
Note: No-observed-effect level - 10,000 ppm

Further information : Note: Excessive exposure may cause central nervous system effects including drowsiness and dizziness. Cardiac Sensitization (dog): No observed effect concentration - 10000 ppm.

12. ECOLOGICAL INFORMATION

Aquatic and ecotoxicity

Toxicity to fish : LC50: 38 mg/l
Exposure time: 96 h
Species: Oncorhynchus mykiss (rainbow trout)
Method: OECD Test Guideline 203

: NOEC: 89.7 mg/l
Exposure time: 192 h
Species: Embryos of Zebra fish (Danio rerio) not older than 8h after fertilization
Method: The guidelines for the testing of chemicals 212: Fish, short-term toxicity test on embryos and sac-fry stages
Note: mortality

Toxicity to daphnia and other aquatic invertebrates : Immobilization
EC50: 82 mg/l
Exposure time: 48 h
Species: Daphnia magna (Water flea)

Method: OECD Test Guideline 202

Toxicity to algae	:	Growth inhibition EC50: > 215 mg/l Exposure time: 72 h Species: Pseudokirchneriella subcapitata (green algae) Method: OECD Test Guideline 201
	:	Growth rate NOEC: 115 mg/l Exposure time: 72 h Species: Pseudokirchneriella subcapitata (green algae) Method: OECD Test Guideline 201
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: 20.1 mg/l Exposure time: 21 d End point: Reproduction Test Species: Daphnia magna (Water flea) Method: The guidelines for the testing of chemicals 211: Daphnia magna reproduction test

Persistence and degradability

Biodegradability	:	Result: Not readily biodegradable. Value: 0 % Method: OECD 301 D

Bioaccumulative potential

Bioaccumulation	:	Note: Due to the distribution coefficient n-octanol/water, accumulation in organisms is not expected.

Ecotoxicology Assessment

Results of PBT assessment
This substance is not considered to be very persistent and very bioaccumulating (vPvB)., This substance is not considered to be persistent, bioaccumulating and toxic (PBT).

13. DISPOSAL CONSIDERATIONS

CN HW China. National Catalogue of Hazardous	:	Hazardous waste

Wastes

Disposal methods	:	In accordance with local and national regulations.
Disposal methods	:	Chemical residues in raw materials by the "National List of Hazardous Wastes" of the regulations, a hazardous waste, pollutants are according to Chinese state or local laws and regulations of the waste classification. According to our law requires specialized units to deal with commission, usually the final disposal of incineration or other legal means to dispose of.

14. TRANSPORT INFORMATION

CNDG

UN/ID No.	:	UN 3163
Description of the goods	:	LIQUEFIED GAS, N.O.S. (TRANS-1-CHLORO-3,3,3-TRIFLUOROPROPENE)
Class	:	2.2
Labels	:	2.2

IATA

UN/ID No.	:	UN 3163
Description of the goods	:	Liquefied gas, n.o.s. (Trans-1-Chloro-3,3,3-trifluoropropene)
Class	:	2.2
Labels	:	2.2
Packing instruction (cargo aircraft)	:	200
Packing instruction (passenger aircraft)	:	200

IMDG

UN/ID No.	:	UN 3163
Description of the goods	:	LIQUEFIED GAS, N.O.S. (TRANS-1-CHLORO-3,3,3-TRIFLUOROPROPENE)
Class	:	2.2
Labels	:	2.2
EmS Number 1	:	F-C
EmS Number 2	:	S-V
Marine pollutant	:	no

15. REGULATORY INFORMATION

National regulatory information

: Refer to the following regulations: Dangerous Chemicals Management Ordinance, the State Council Order No. 591; GB16483-2008 MSDS content and order of items; GB-T 17519-2013 Material Safety Data Sheet written rules; GB12268-2012 Name List of dangerous goods; GB6944-2012 and the name of dangerous goods classification number; GB13690-2009 chemical formula for risk classification and General; GBZ 2.1-2019 workplace hazards Part 1 occupational exposure limit chemical hazards; GB30000.2~29-2013 classification of chemicals, warning labels and warning statements of safety norms

Other international regulations

Notification status

US. Toxic Substances Control Act	:	On TSCA Inventory
Australia. Industrial Chemical (Notification and Assessment) Act	:	On the inventory, or in compliance with the inventory
Canada. Canadian Environmental Protection Act (CEPA). Domestic Substances List (DSL)	:	All components of this product are on the Canadian DSL
Japan. Kashin-Hou Law List	:	On the inventory, or in compliance with the inventory
Korea. Existing Chemicals Inventory (KECI)	:	On the inventory, or in compliance with the inventory
Philippines. Inventory of Chemicals and Chemical Substances (PICCS)	:	Not in compliance with the inventory
China. Inventory of Existing Chemical Substances	:	On the inventory, or in compliance with the inventory

(IECSC)

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand	:	On the inventory, or in compliance with the inventory

16. OTHER INFORMATION

		HMIS III	NFPA
Health hazard	:	2	2
Flammability	:	0	0
Physical Hazard	:	0	
Instability	:		0

Hazard rating and rating systems (e.g. HMIS® III, NFPA): This information is intended solely for the use of individuals trained in the particular system.

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. Final determination of suitability of any material is the sole responsibility of the user. This information should not constitute a guarantee for any specific product properties.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.
Previous Issue Date: 05/01/2019
Prepared by Honeywell Performance Materials and Technologies Product Stewardship Group